Addressing Mechanisms and Heterogeneity in Change with Usable Tools

Ivan Jacob Agaloos Pesigan (Jek)
ijapesigan@psu.edu

Postdoctoral Fellow
Prevention and Methodology Training Program (PAMT)
Edna Bennett Pierce Prevention Research Center

The Pennsylvania State University

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https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk

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- Good morning, and thank you for the opportunity to be here. I am Ivan Jacob Agaloos Pesiga, Jek for short.
- At the center of my research program is a simple goal: to understand **dynamic processes** in *development*, *family life*, and *health* in ways that are **substantively meaningful** and **methodologically usable**.
- In particular, I study *how change unfolds over time*, *how people differ in those processes*, and *how we can build tools that help researchers study those questions well*.
- What draws me to HDFS is that these questions are inherently **contextual** and **multilevel**, showing up in *individuals*, *relationships*, *families*, and *everyday* settings.
- Today, I will walk through three recent studies that reflect that agenda.
- Across them, the common thread is **methodology in service of substantive questions**.
- I will end with a few future directions I would be especially excited to pursue at Penn State HDFS.

Training and mentorship

The Ateneo de Manila University

University of Macau

The Pennsylvania State University

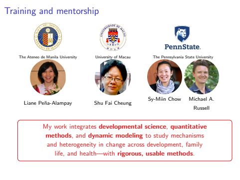
Liane Peña-Alampay

Shu Fai Cheung

Sy-Miin Chow

Michael A. Russell

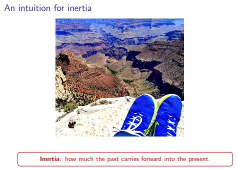
My work integrates **developmental science**, **quantitative methods**, and **dynamic modeling** to study mechanisms and heterogeneity in change across development, family life, and health—with **rigorous, usable methods**.



- Before I get into the studies, I want to briefly situate the kind of work I do and how I was trained for it.
- At Ateneo de Manila University, I built my foundation in **psychology**, with strong grounding in *development*, *context*, and *teaching*. My mentor there, Liane Peña-Alampay, is a Penn State HDFS alumna, so that connection makes being here especially meaningful to me.
- At Ateneo, I also had the chance to **teach courses** in statistics, psychometrics, and psychology. That experience shaped how I think about methods: not as ends in themselves, but as tools that should be explainable, useful, and connected to substantive questions.
- At the University of Macau, working with Shu Fai Cheung, I deepened my training in **structural equation modeling**, **meta-analysis**, and **statistical computation**.
- Then here at Penn State, through PAMT and my work with Sy-Miin Chow and Michael Russell, I extended that foundation into **dynamic models** and **intensive longitudinal data**, especially in applications tied to *health* and *behavior* in everyday life.

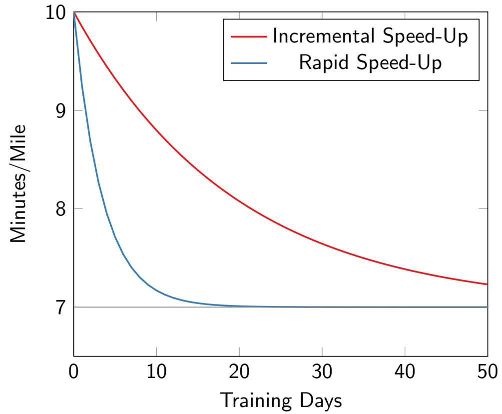
An intuition for inertia

Inertia: how much the past carries forward into the present.

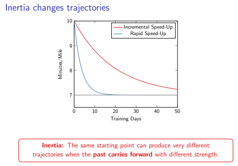


- This photo is from the Grand Canyon when I was in graduate school. Around that time, I was also doing a lot of running, and running gave me a simple way to think about dynamics.
- In training, your pace today is often shaped by what happened the day before. A hard run can carry forward into the next day through fatigue, while steady training can carry forward through adaptation and momentum.
- A lot of the processes we care about in HDFS work the same way: **they do not reset from one moment to the next**. What happened yesterday, or even a few hours ago, can still shape what happens today.
- That is what I mean by **inertia**: how much the past carries forward into the present.

Inertia changes trajectories

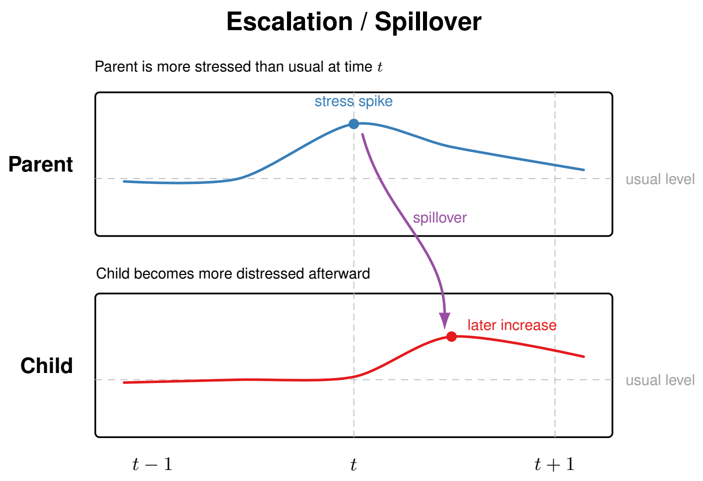


Inertia: The same starting point can produce very different trajectories when the **past carries forward** with different strength.

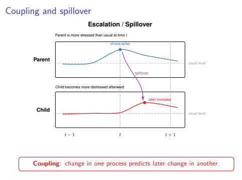


- Here both runners start at the same place, but their trajectories diverge over time.
- The difference is not where they began. The difference is how strongly earlier states carry forward.
- When inertia is stronger (red curve), the past keeps exerting influence for longer, so the trajectory changes more slowly.
- When inertia is weaker (blue curve), the system adjusts more quickly and settles back faster.
- That is the key intuition I want to carry into the rest of the talk: dynamic processes are shaped not only by level, but by persistence.
- In HDFS, that kind of persistence shows up in affective recovery, stress carryover, sleep, health behavior, and many other daily-life processes.
- But dynamic processes are not only self-persistent. They can also shape one another.

Coupling and spillover



Coupling: change in one process predicts later change in another.



- Here, a parent's stress spike is followed by greater child distress. That is the basic idea of **coupling**: change in one process can predict later change in another.
- So now we move from inertia within a single process to spillover across connected processes.
- Those two ideas—**inertia** and **coupling**—set up the rest of the talk.

Study 1

Psychological Methods

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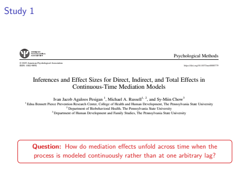
Inferences and Effect Sizes for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

Ivan Jacob Agaloos Pesigan [1], Michael A. Russell[1, 2], and Sy-Miin Chow[3]
[1] Edna Bennett Pierce Prevention Research Center, College of Health and Human Development, The Pennsylvania State University
[2] Department of Biobehavioral Health, The Pennsylvania State University
[3] Department of Human Development and Family Studies, The Pennsylvania State University

Question: How do mediation effects unfold across time when the process is modeled continuously rather than at one arbitrary lag?



- Once we allow processes to persist and to shape one another, a natural next question is whether one process affects another **through** an intermediate pathway.
- So instead of asking only whether stress predicts later mood, we can ask whether part of that effect is transmitted through something like coping.
- But as soon as we ask that question over time, the answer can depend on **when** we look: the same indirect pathway may appear stronger, weaker, or even missed entirely depending on the spacing of the observations.
- That is why Study 1 moves to **continuous time** - to define **direct**, **indirect**, and **total** effects as processes that unfold across time itself, rather than as results tied to one arbitrary observation interval.

Why mediation has been a recurring part of my work

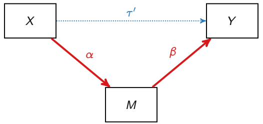
- **Substantive** (e.g.; Bernardo, Wang, Pesigan, & Yeung, 2017; Pesigan, Luyckx, & Alampay, 2014; Zhang, Ku, Wu, Yu, & Pesigan, 2020)
- **Methodological** (e.g.; Cheung & Pesigan, 2023b; Cheung, Pesigan, & Vong, 2023; Pesigan & Cheung, 2020, 2024; Pesigan, Sun, & Cheung, 2023).
  - https://CRAN.R-project.org/package=semmcci
  - https://CRAN.R-project.org/package=semlbci
  - https://CRAN.R-project.org/package=betaDelta
  - https://CRAN.R-project.org/package=betaDSandwich

Study 1 extends that line of work into continuous time by adding **standardized effect sizes**, principled **uncertainty quantification**, and an open-source tool in `cTMed`.

Why mediation has been a recurring part of my work
Study 1 extends that line of work into continuous time by adding standardized effect sizes, principled uncertainty quantification, and an open-source tool in cTMed.

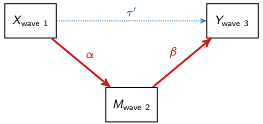
- Mediation has been a recurring part of my work from both substantive and methodological angles.
- On the substantive side, I have used mediation to study mechanisms in applied problems, including alcohol use and related health behaviors.
- On the methods side, I have worked on inference for indirect effects and related quantities, including confidence intervals, bootstrap and Monte Carlo approaches, and standardized effect-size inference.
- Study 1 extends that line of work into continuous time.
- Prior work showed how these effects unfold across time; my contribution was to make their interpretation and inference more usable.

Why this problem matters



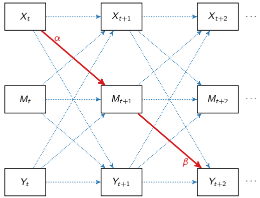
Cross-sectional

One snapshot



Three-wave

Coarse temporal ordering

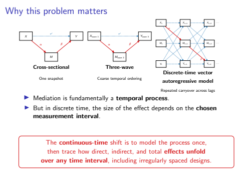


Discrete-time vector autoregressive model

Repeated carryover across lags

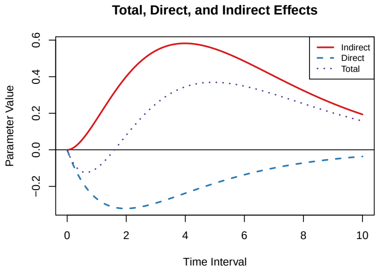
- Mediation is fundamentally a **temporal process**.
- But in discrete time, the size of the effect depends on the **chosen measurement interval**.

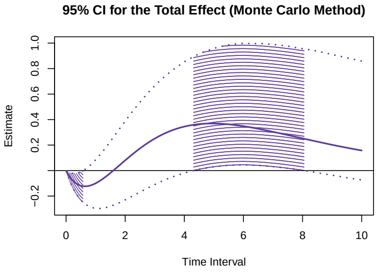
The **continuous-time** shift is to model the process once, then trace how direct, indirect, and total **effects unfold over any time interval**, including irregularly spaced designs.



- This slide is really about temporal resolution.
- A cross-sectional mediation model is like a single photograph: one snapshot of a process that is actually unfolding over time.
- A three-wave model gives us a few still images, so we get some ordering, but only very coarsely.
- A discrete-time autoregressive model gets us closer, but it is still like sampling the process at selected frames. What we see depends on which measurement interval we happened to choose.
- The continuous-time shift is to model the motion itself rather than just a few snapshots of it.
- That lets us trace how the direct, indirect, and total effects unfold over any time interval, including irregularly spaced designs.
- For HDFS researchers, that matters because mechanisms in development, family process, and health behavior rarely operate at one arbitrary lag.

Novel contributions





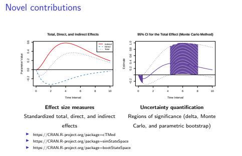
Effect size measures

Standardized total, direct, and indirect effects

Uncertainty quantification

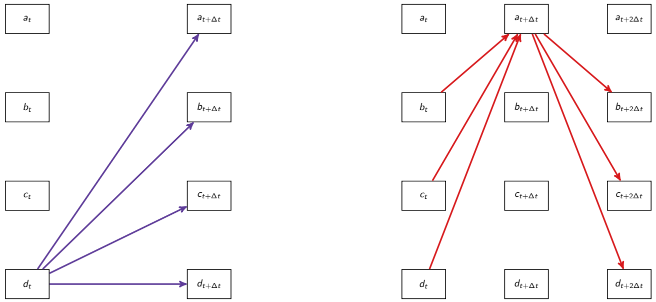
Regions of significance (delta, Monte Carlo, and parametric bootstrap)

- https://CRAN.R-project.org/package=cTMed
- https://CRAN.R-project.org/package=simStateSpace
- https://CRAN.R-project.org/package=bootStateSpace



- Start with the left panel. Think of **stress** affecting later **mood**, partly through **coping**.
- The key point is that in continuous time, the direct, indirect, and total effects are not tied to one lag; we can see how the effect of stress on mood unfolds over different intervals.
- Early on, the effect of stress on mood is mostly **direct**. A bit later—around 2 to 4 on the x-axis—more of that effect is carried **indirectly** through coping.
- The **total** effect of stress on mood peaks a little later and then tapers. These are **standardized** effects, so they are interpretable much like standardized regression coefficients.
- Now look at the right panel. This shows **uncertainty quantification** for the total effect using the Monte Carlo method.
- The shaded region marks the intervals where the total effect of stress on mood is statistically different from zero—roughly around 4 to 8. Outside that range, the interval includes zero.
- More broadly, I developed delta, Monte Carlo, and parametric bootstrap approaches for this problem, with Monte Carlo offering a useful middle ground between speed and computational cost.
- And these quantities and plots can be generated directly using the open-source `cTMed` package.

One direction this line of work opens up



Total effect centrality

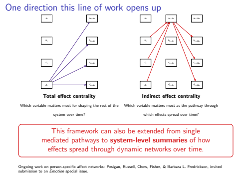
Which variable matters most for shaping the rest of the system over time?

Indirect effect centrality

Which variable matters most as the pathway through which effects spread over time?

This framework can also be extended from single mediated pathways to **system-level summaries** of how effects spread through dynamic networks over time.

Ongoing work on person-specific affect networks: Pesigan, Russell, Chow, Fisher, & Barbara L. Fredrickson, invited submission to an *Emotion* special issue.



- One reason I am excited about this work is that it opens up a broader systems perspective.
- Study 1 focused on mediation one pathway at a time, but many substantive questions are really about how an **entire system reorganizes** over time.
- That is the idea behind an ongoing project on **person-specific affect networks during meditation training**, across baseline, intervention, and follow-up.
- Here, **total-effect centrality** asks which emotion has the *broadest downstream* influence, and **indirect-effect centrality** asks which emotion serves as a key *conduit* through which effects spread.
- Because the model is continuous-time, we can examine those roles across **multiple timescales**—from short-term ripple effects to longer-horizon accumulation.
- And because this is an intervention study, we can link that reorganization to **meditation practice dose** and to **follow-up affective functioning and well-being**.
- So the broader payoff of Study 1 is a framework for turning complex dynamics into interpretable summaries that show **which variables have the widest downstream influence and which variables serve as the main pathways through which effects spread over time**.
- And once we can describe dynamic systems that way, the next question is whether those dynamics vary meaningfully across individuals.

Study 2

AMERICAN PSYCHOLOGICAL ASSOCIATION

Psychology of Addictive Behaviors

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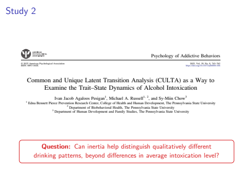
https://doi.org/10.1037/adb0001106

Common and Unique Latent Transition Analysis (CULTA) as a Way to Examine the Trait–State Dynamics of Alcohol Intoxication

Ivan Jacob Agaloos Pesigan[1], Michael A. Russell[1, 2], and Sy-Miin Chow[3]
[1] Edna Bennett Pierce Prevention Research Center, College of Health and Human Development, The Pennsylvania State University
[2] Department of Biobehavioral Health, The Pennsylvania State University
[3] Department of Human Development and Family Studies, The Pennsylvania State University

Question: Can inertia help distinguish qualitatively different drinking patterns, beyond differences in average intoxication level?



- That is the question behind Study 2: can inertia help distinguish qualitatively different drinking patterns, beyond differences in average intoxication level?
- Put differently, people may differ not only in how much they drink, but in how strongly intoxication carries forward across days.

Why this substantive problem demanded a new model

Why Study 2 is exciting

- **Substantive grounding:** Alcohol and addiction have been a consistent part of my applied work (e.g. Yu, Pesigan, Zhang, & Wu, 2019; Yu, Wu, & Pesigan, 2015; Zamboanga, Iwamoto, Pesigan, & Tomaso, 2015; Zamboanga, Pesigan, Tomaso, Schwartz, et al., 2015; Zhang, Ku, Wu, Yu, & Pesigan, 2020; Zhang, Pesigan, Kahler, Yip, et al., 2019).
- **Substantive question:** How do risky drinking patterns persist or change across days?
- **Methodological challenge:** Transdermal alcohol concentration (TAC) features are multidimensional, partly shared, partly feature-specific, and dynamic over time.

Substantive grounding in addiction research,
together with the quantitative expertise to build
new models when the science demands them.

Why this substantive problem demanded a new model
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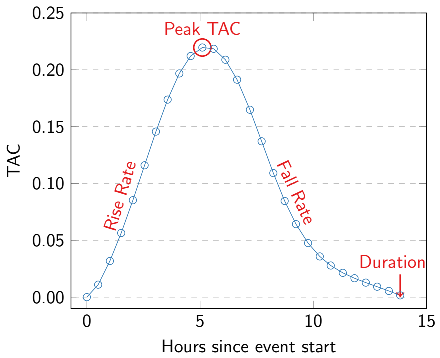
Substantive question: How do risky drinking patterns persist or change across days?
Methodological challenge: Transdermal alcohol concentration (TAC) features are multidimensional, partly shared, partly feature-specific, and dynamic over time.
Substantive grounding in addiction research, together with the quantitative expertise to build new models when the science demands them.

- This study is a natural next step for me.
- Addiction and alcohol have been a consistent substantive thread in my work, but Study 2 is where that background meets the kind of quantitative contribution I want to make in HDFS.
- TAC sensors give us richer and more objective data, but they also create a harder modeling problem: the features are partly shared, partly feature-specific, and changing across days.
- That is exactly the setting that motivated CULTA.

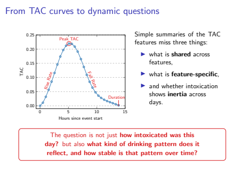
From TAC curves to dynamic questions



Simple summaries of the TAC features miss three things:

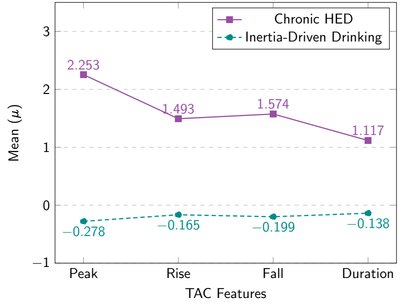
- what is **shared** across features,
- what is **feature-specific**,
- and whether intoxication shows **inertia** across days.

The question is not just **how intoxicated was this day?** but also **what kind of drinking pattern does it reflect, and how stable is that pattern over time?**

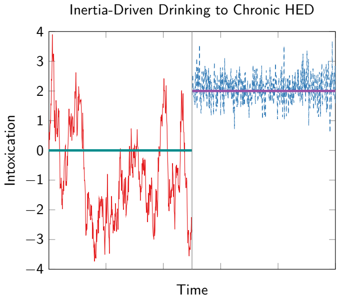


- Here, the key data are **repeated day-specific TAC curves**: across the monitoring period, each participant can contribute multiple drinking-day curves, and from each curve we extract features like peak, rise rate, fall rate, and duration.
- But once we do that, a new question appears: are we looking at one common intoxication process, feature-specific expressions of intoxication, or both?
- And beyond that, does intoxication carry over across days?
- So the substantive question becomes not just how intoxicated this day was, but what kind of drinking pattern it reflects and how stable that pattern is over time.
- That is exactly the problem CULTA was designed to solve.

Profiles differed in mean intoxication and inertia



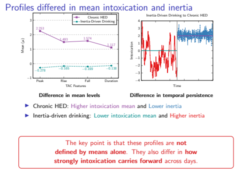
Difference in mean levels



Difference in temporal persistence

- Chronic HED: Higher intoxication mean and Lower inertia
- Inertia-driven drinking: Lower intoxication mean and Higher inertia

The key point is that these profiles are **not defined by means alone**. They also differ in **how strongly intoxication carries forward** across days.



- CULTA answers that question by separating overall intoxication level from temporal persistence.
- Two profiles emerged, and the key point is how they differ.
- **Chronic HED** shows higher mean intoxication and lower inertia, which is more consistent with an entrenched heavy-drinking pattern.
- **Inertia-driven drinking** shows lower mean intoxication but stronger carryover across days, suggesting a pattern where recent history—and likely context—still matters for what happens next.
- So this is not just a classification result. It shows that risky drinking patterns differ in both level and persistence.

AUD risk predicts getting stuck in chronic HED

Start in chronic HED	Stay in chronic HED	Shift into chronic HED
No risk → Risk → Likely AUD	No risk → Risk → Likely AUD	No risk → Risk → Likely AUD
0.03 → 0.07 → 0.15	0.21 → 0.30 → 0.40	0.03 → 0.06 → 0.10

CULTA does not just classify drinking days; it helps identify **who is likely to get stuck in riskier dynamic patterns**.

AUD risk predicts getting stuck in chronic HED
Start in chronic HED Stay in chronic HED Shift into chronic HED
No risk → Risk → Likely AUD No risk → Risk → Likely AUD No risk → Risk → Likely AUD
0.03 → 0.07 → 0.15 0.21 → 0.30 → 0.40 0.03 → 0.06 → 0.10
CULTA does not just classify drinking days; it helps identify who is likely to get stuck in riskier dynamic patterns.

- AUD risk helps identify who is more likely to start in, remain in, or shift into the chronic HED profile.
- The main pattern is simple: as AUD risk increases, so does the probability of getting stuck in the chronic pattern.
- So Study 2 shows that risky drinking patterns are not defined by average intoxication alone, but also by persistence and who gets stuck in maladaptive dynamics over time.
- And once we start treating these dynamic signatures as meaningful individual differences, the next question is how to synthesize them at the population level without averaging them away.
- That is exactly the problem I take up in Study 3.

Study 3

Under review at *Multivariate Behavioral Research*

From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

Ivan Jacob Agaloos Pesigan[1], Michael A. Russell[1, 2], and Sy-Miin Chow[3]
[1]Edna Bennett Pierce Prevention Research Center, The Pennsylvania State University
[2]Department of Biobehavioral Health, The Pennsylvania State University
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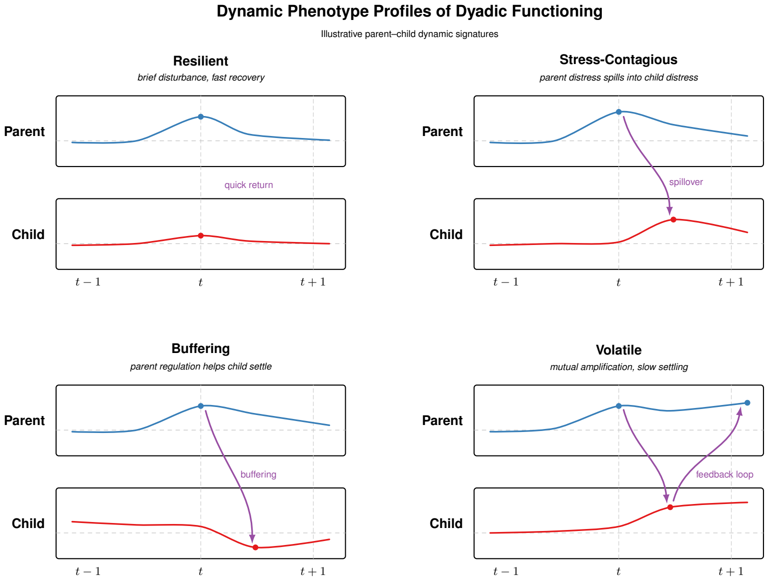
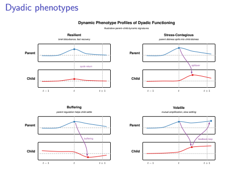
Question: How can we draw **population-level** conclusions from **person-specific dynamics** without averaging away the very individual differences we care about?

Study 3
From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models
Question: How can we draw population-level conclusions from person-specific dynamics without averaging away the very individual differences we care about?

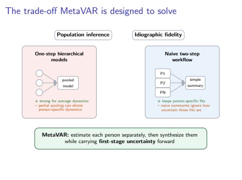
- Study 3 asks how to preserve **person-specific dynamics** while still making valid **population-level** inferences.
- For me, this is not only a statistical problem.
- It is the problem behind many HDFS questions, because we often care about **dynamic parameters** that capture regulation, adaptation, spillover, and risk in ways that **differ meaningfully across people, dyads, and families**.
- So the question is how to synthesize those individual dynamic signatures without averaging away the heterogeneity that makes them substantively important.

Dyadic phenotypes

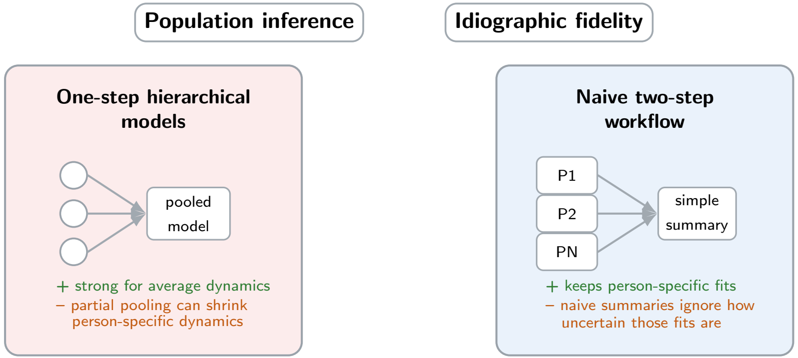




- Before I show the mechanics of MetaVAR, I want to show the kind of scientific target I care about.
- Imagine estimating a dynamic signature for each parent–child dyad: some dyads recover quickly after disturbance, some show stress contagion, some show buffering, and some show mutual amplification.
- Those are not just abstract parameters. They are candidate dynamic phenotypes of dyadic regulation and functioning.
- And once we can estimate those signatures well, the next question is **how to synthesize them without flattening the very heterogeneity that makes them interesting**.
- That is the problem MetaVAR is designed to solve.



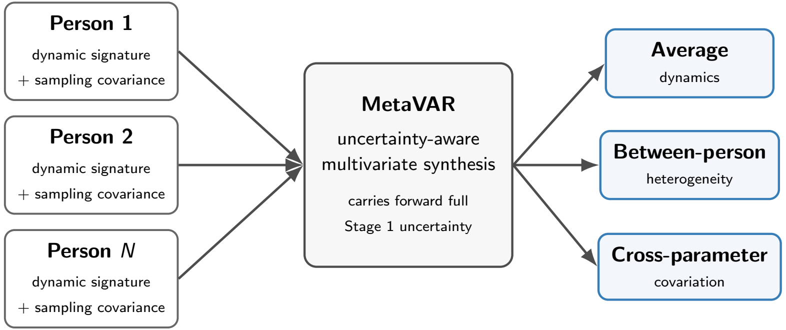
The trade-off MetaVAR is designed to solve



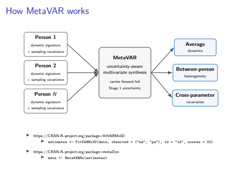
MetaVAR: estimate each person separately, then synthesize them while carrying **first-stage uncertainty** forward

- There is a real trade-off here.
- One-step hierarchical models are attractive because they estimate population averages and individual differences together. But the cost is that person-specific dynamics can get pulled toward the population mean.
- At the other extreme, we can fit a separate model per person and summarize afterward. That preserves the idiographic fits and scales well, but it can treat all person-level estimates as if they were equally precise.
- MetaVAR is the middle ground: keep the person-specific models, but carry their uncertainty into the second stage.
- In HDFS terms, this matters whenever we care not only about the average person, dyad, or family, but also about how much meaningful variation there is around that average.

How MetaVAR works

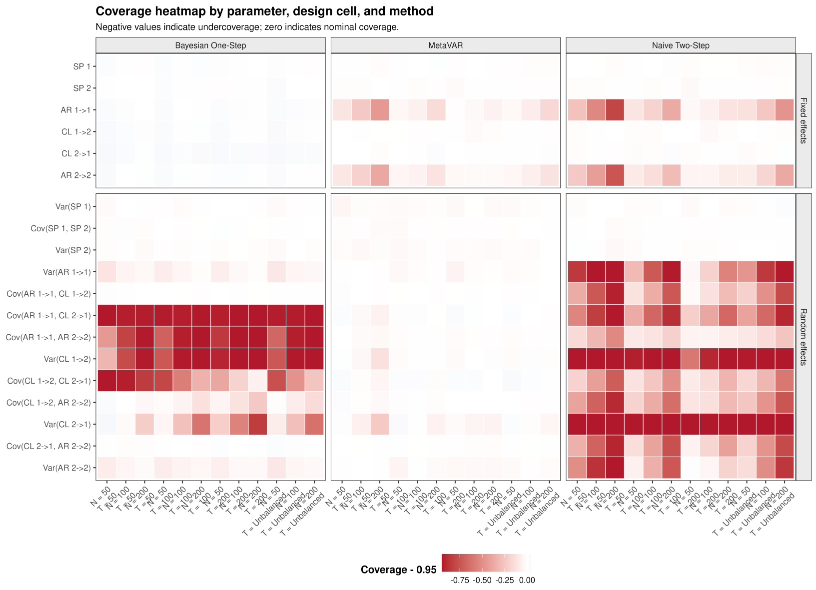


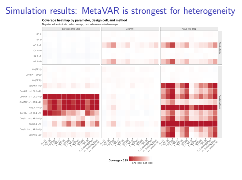
- https://CRAN.R-project.org/package=fitVARMxID
  - `estimates <- FitVARMxID(data, observed = ("na", "pa"), id = "id", ncores = 32)`
- https://CRAN.R-project.org/package=metaDyn
  - `meta <- MetaVARMx(estimates)`



- Stage 1 is person-specific estimation: each individual gets an idiographic model, which yields a dynamic signature and its sampling covariance matrix.
- Stage 2 is MetaVAR: instead of treating those person-specific estimates as error-free, the method synthesizes both the signatures and their uncertainty.
- That gives us valid population-level inference on average dynamics, between-person heterogeneity, and covariation among dynamic parameters.

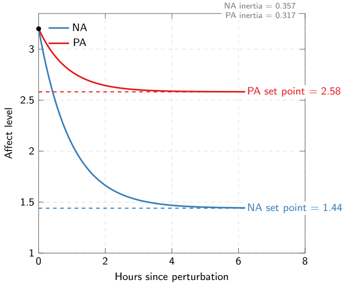
Simulation results: MetaVAR is strongest for heterogeneity

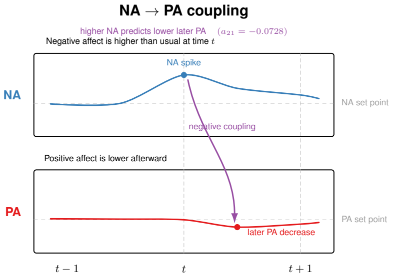


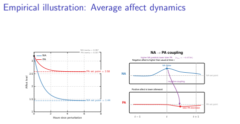


- The simulation clarifies where MetaVAR is especially useful.
- Its clearest advantage is when the scientific target is **heterogeneity**: how much people differ, and how well we calibrate uncertainty about those differences.
- For population averages, the Bayesian multilevel comparator remained competitive.
- So I see MetaVAR not as a universal replacement, but as a strong option when we want to preserve idiographic estimation while still drawing valid population-level inferences about heterogeneity.

Empirical illustration: Average affect dynamics

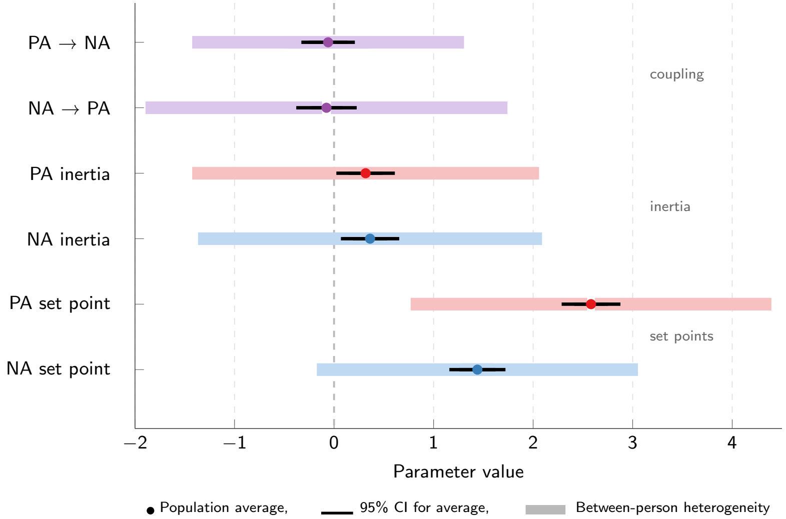


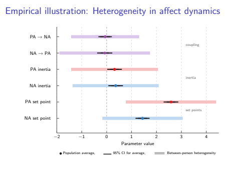




- So let me now show what that looks like in real data.
- I will start with the population-average affect system, not because the mean is the whole story, but because it gives us an interpretable baseline.
- On average, people show higher positive affect than negative affect, and both processes show positive inertia.
- I am also showing one representative cross-lagged result here: higher negative affect predicts lower later positive affect.
- The companion result goes in the same direction for positive affect predicting lower later negative affect, so the average system reflects **reciprocal inhibitory coupling**.
- Then, in the next slide, I will show the part MetaVAR is especially good at: how much people differ around that average.

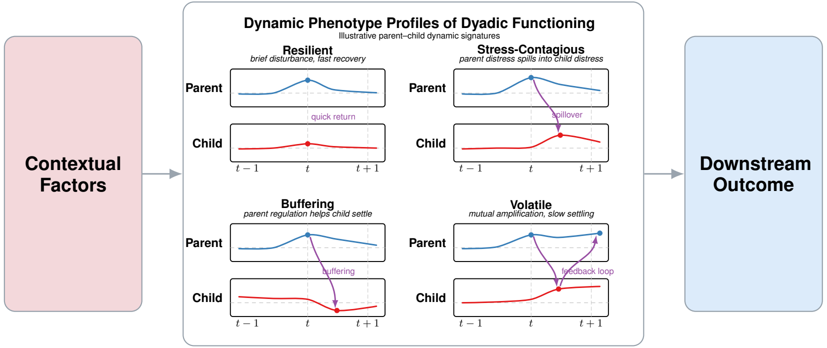
Empirical illustration: Heterogeneity in affect dynamics



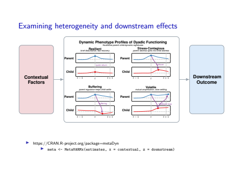


- The more important result is not only where the average lies, but how much people differ around it.
- The dot is the population average, the thin interval is the 95% confidence interval for that average, and the wider band shows approximate between-person heterogeneity around it.
- That wider band comes from the random-effects variance for each parameter, so it reflects population spread rather than uncertainty about the mean.
- So people differ meaningfully in their set points, in how persistent affect is, and in how strongly negative and positive affect influence one another over time.
- The payoff is that heterogeneity stops being a nuisance term and becomes a substantive result: people differ in their set points, persistence, coupling, and co-fluctuation, and those differences are exactly what we often care about in HDFS.

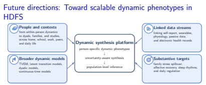
Examining heterogeneity and downstream effects



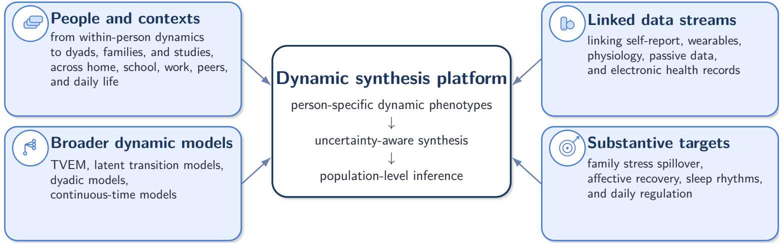
- https://CRAN.R-project.org/package=metaDyn
  - `meta <- MetaVARMx(estimates, x = contextual, z = downstream)`



- This slide shows why MetaVAR matters beyond a single methods paper.
- In a broader SEM framework, person-specific dynamic signatures can be modeled as outcomes of contextual factors and as predictors of later outcomes.
- So instead of stopping at “people differ,” we can ask what predicts those differences and why they matter downstream.
- That is the bridge to the next slide: toward dynamic phenotypes in HDFS.

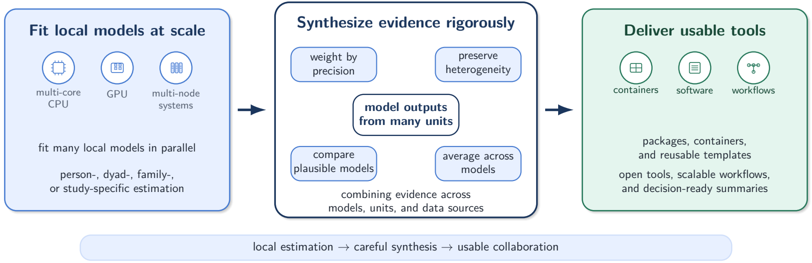


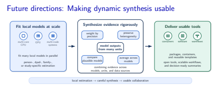
Future directions: Toward scalable dynamic phenotypes in HDFS



- This slide is about the 5- to 10-year arc of my research program.
- A central challenge is scale: many existing dynamic methods become harder to use as we add more variables, more levels, and richer data sources.
- My goal is to build a scalable synthesis framework for dynamic phenotypes: estimate them locally, carry uncertainty forward, and combine them rigorously.
- The long-term aim is methods that scale with HDFS science without losing rigor, interpretability, or usability.

Future directions: Making dynamic synthesis usable





- This slide is about infrastructure, in three senses.
- First, the hardware backend: CPUs, GPUs, and HPC nodes that let us estimate many local dynamic models in parallel.
- Second, careful synthesis: translating statistical ideas into algorithms that combine those local models rigorously while preserving uncertainty and heterogeneity.
- Third, usable tools: packages, containers, and workflows that put those methods into forms collaborators and students can actually use.
- That is the kind of infrastructure I would want to bring to the department—infrastructure that supports collaboration, training, and reproducible quantitative work.
- And ultimately, it is all in service of the same goal: studying dynamic processes in development, family life, and health in ways that are substantively meaningful and methodologically usable.

Matri meae carissimae, VAP.

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- Thank you very much for your attention.
- I have tried to show a research program focused on dynamic processes, meaningful heterogeneity, and usable tools for studying development, family life, and health.
- That is why HDFS feels like such a strong fit for me, and I would be excited to contribute here through methods development, collaboration, and student training.
- Thank you, and I would be happy to take questions.

Addressing Mechanisms and Heterogeneity in Change with Usable Tools

Ivan Jacob Agaloos Pesigan (Jek)
ijapesigan@psu.edu

Postdoctoral Fellow
Prevention and Methodology Training Program (PAMT)
Edna Bennett Pierce Prevention Research Center

The Pennsylvania State University

April 20, 2026

https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk

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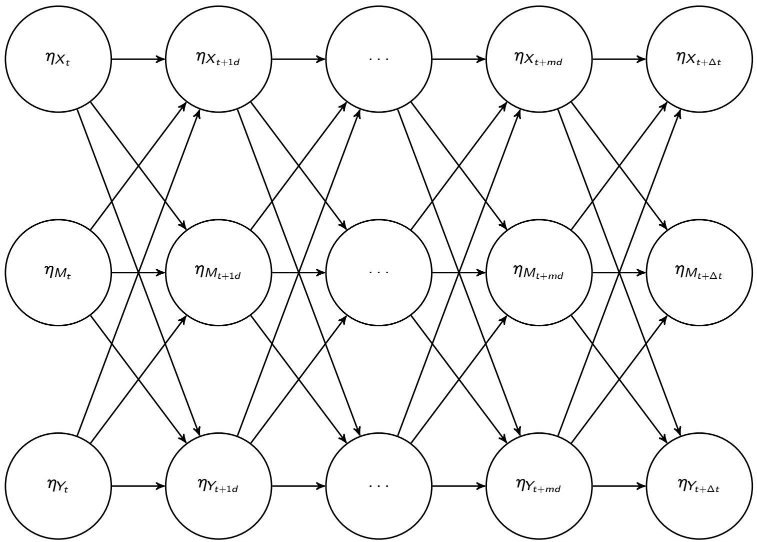
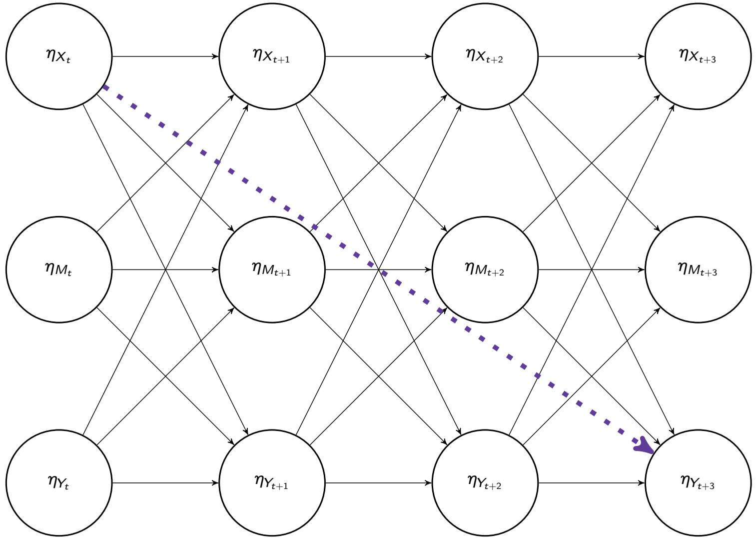
Extra slides

Continuous-Time Vector Autoregressive Model (CT-VAR: Drift Matrix $\mathbf{\Phi}$)



Continuous-Time Vector Autoregressive Model (CT-VAR: Drift Matrix $\mathbf{\Phi}$)

Total Effect Decomposition (Deboeck & Preacher, 2015; Pesigan, Russell, & Chow, 2025b; Ryan & Hamaker, 2021)



2026-04-21

Addressing Mechanisms and Heterogeneity in Change with Usable Tools
└─Extra slides
  └─Total Effect Decomposition (Deboeck & Preacher, 2015; Pesigan, Russell, & Chow, 2025b; Ryan & Hamaker, 2021)

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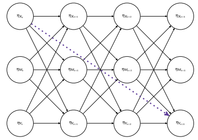
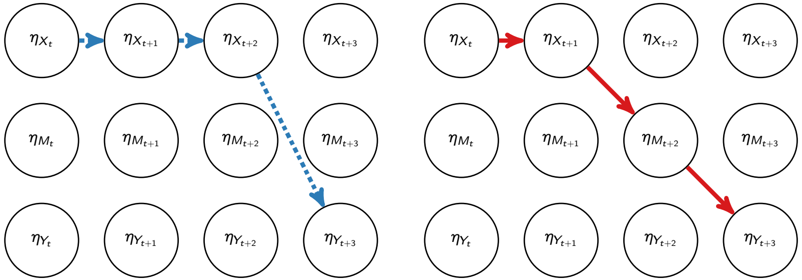


Total Effect Decomposition

Blue $=$ direct effect $(\beta_{11}\beta_{11}\beta_{31})$. Red $=$ indirect effect $(\beta_{11}\beta_{21}\beta_{32})$.



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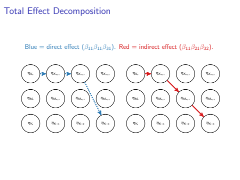
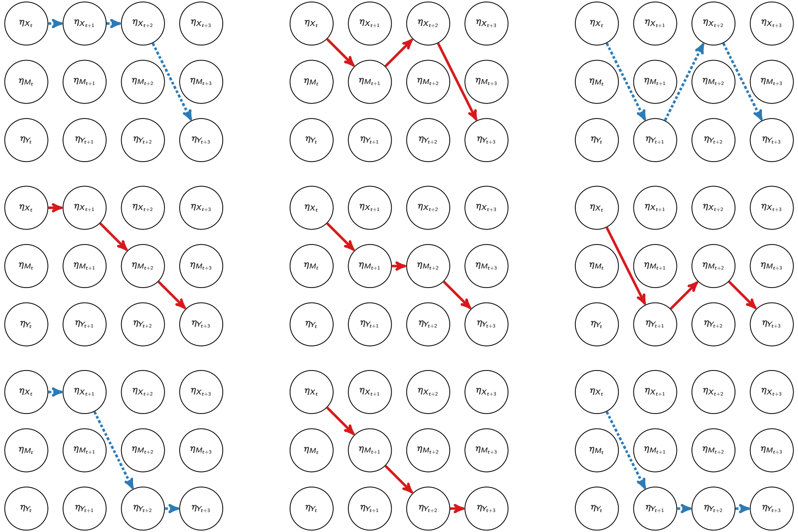
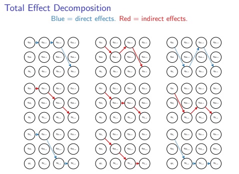


Total Effect Decomposition

Blue = direct effects. Red = indirect effects.





Total Effect Decomposition

Total, Direct, and Indirect Effects:

$$\text{Total} = \exp\left(\Delta t \boldsymbol{\Phi}\right) = \boldsymbol{\beta}_{\Delta t} \tag{1}$$

$$\text{Direct} = \exp\left(\Delta t \mathbf{D_m} \boldsymbol{\Phi} \mathbf{D_m}\right) \tag{2}$$

$$\text{Indirect} = \text{Total} - \text{Direct}. \tag{3}$$

Standardized Effects: Effect Size Measures

$$\text{Total}^{*}_{i,j} = \text{Total}_{i,j}\left(\frac{\sigma_{x_j}}{\sigma_{y_i}}\right), \tag{4}$$

$$\text{Direct}^{*}_{i,j} = \text{Direct}_{i,j}\left(\frac{\sigma_{x_j}}{\sigma_{y_i}}\right), \qquad \text{and} \tag{5}$$

$$\text{Indirect}^{*}_{i,j} = \text{Total}^{*}_{i,j} - \text{Direct}^{*}_{i,j} \tag{6}$$

Total Effect Decomposition
Total, Direct, and Indirect Effects:
Standardized Effects: Effect Size Measures

CT-VAR (Chow et al., 2023; Deboeck & Preacher, 2015)

Measurement component:

$$\mathbf{y}_{i,t_m} = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta}_{i,t_m} + \boldsymbol{\varepsilon}_{i,t_m}, \boldsymbol{\varepsilon}_{i,t_m} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Theta}\right) \tag{7}$$

Dynamic component:

$$\mathrm{d}\boldsymbol{\eta}_{i,t} = \left(\boldsymbol{\Phi}\boldsymbol{\eta}_{i,t}\right)\mathrm{d}t + \boldsymbol{\Sigma}^{\frac{1}{2}}\mathrm{d}\mathbf{W}_{i,t} \tag{8}$$

Three-variable CT-VAR mediation model:

$$\mathbf{y} = \begin{pmatrix} X \\ M \\ Y \end{pmatrix}, \quad \text{and} \quad \boldsymbol{\eta} = \begin{pmatrix} \eta_X \\ \eta_M \\ \eta_Y \end{pmatrix} \tag{9}$$

Integral form:

$$\boldsymbol{\eta}_{i,t_{m_i}} = \boldsymbol{\alpha}_{\Delta t_{m_i}} + \boldsymbol{\beta}_{\Delta t_{m_i}}\boldsymbol{\eta}_{i,t_{m_i-1}} + \boldsymbol{\zeta}_{i,t_{m_i}}, \boldsymbol{\zeta}_{i,t_{m_i}} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Psi}_{\Delta t_{m_i}}\right) \tag{10}$$

$$\boldsymbol{\beta}_{\Delta t_{m_i}} = \exp\left(\Delta t\boldsymbol{\Phi}\right) \tag{11}$$

CT-VAR (Chow et al., 2023; Deboeck & Preacher, 2015)
Measurement component:
Dynamic component:
Three-variable CT-VAR mediation model:
Integral form:

Delta Method Confidence Intervals

We define the delta method following the multivariate central limit theorem. Let $\boldsymbol{\theta}$ be $\mathrm{vec}\left(\boldsymbol{\Phi}\right)$, that is, the elements of the drift matrix $\boldsymbol{\Phi}$ in vector form sorted column-wise. Let $\hat{\boldsymbol{\theta}}$ be $\mathrm{vec}\left(\hat{\boldsymbol{\Phi}}\right)$, that is, the estimates of the elements of the drift matrix. The uncertainty around any functions of the estimates for $\hat{\boldsymbol{\theta}}$ is assumed to be characterized as:

$$\sqrt{n}\left(\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right) - \mathbf{g}\left(\boldsymbol{\theta}\right)\right) \xrightarrow{\mathrm{D}} \mathcal{N}\left(0, \mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right) \tag{12}$$

where $\mathbf{g}$ is a vector of functions, $\mathbf{J}$ is the matrix of first-order derivatives of the functions in $\mathbf{g}$ with respect to the elements of $\boldsymbol{\theta}$ and $\boldsymbol{\Gamma}$ is the asymptotic variance-covariance matrix of $\hat{\boldsymbol{\theta}}$. The approximate distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is given by

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right) \approx \mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right), n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right). \tag{13}$$

When $\boldsymbol{\Gamma}$ is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of $\hat{\boldsymbol{\theta}}$, that is, $\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)$ for $n^{-1}\boldsymbol{\Gamma}$.

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Monte Carlo (MC) Method Confidence Intervals

Based on the asymptotic properties of maximum likelihood estimators, we can assume that they are normally distributed around the population parameters

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}\left(\boldsymbol{\theta}, \mathbb{V}\left(\hat{\boldsymbol{\theta}}\right)\right). \tag{14}$$

Using this distributional assumption, a sampling distribution of $\hat{\boldsymbol{\theta}}$ which we refer to as $\hat{\boldsymbol{\theta}}^{*}$ can be generated by replacing the population parameters with sample estimates, that is,

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A sampling distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$, which we refer to as $\mathbf{g}\left(\hat{\boldsymbol{\theta}}^{*}\right)$, can be generated by using the simulated estimates to calculate $\mathbf{g}$. The standard deviations of the simulated estimates are the SEs. In resampling approaches, it is customary to use the percentiles of the generated distribution as CIs instead of the SEs.

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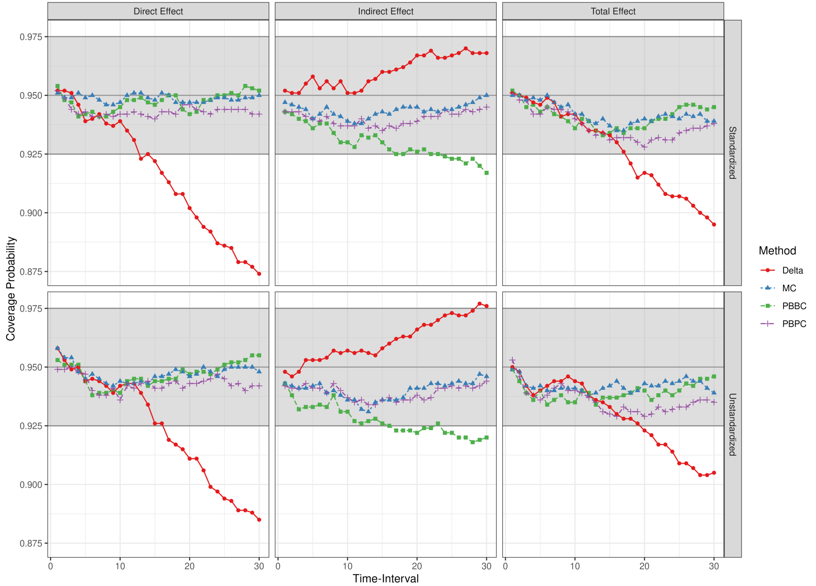
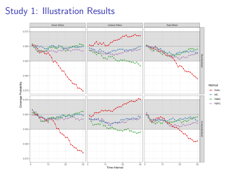
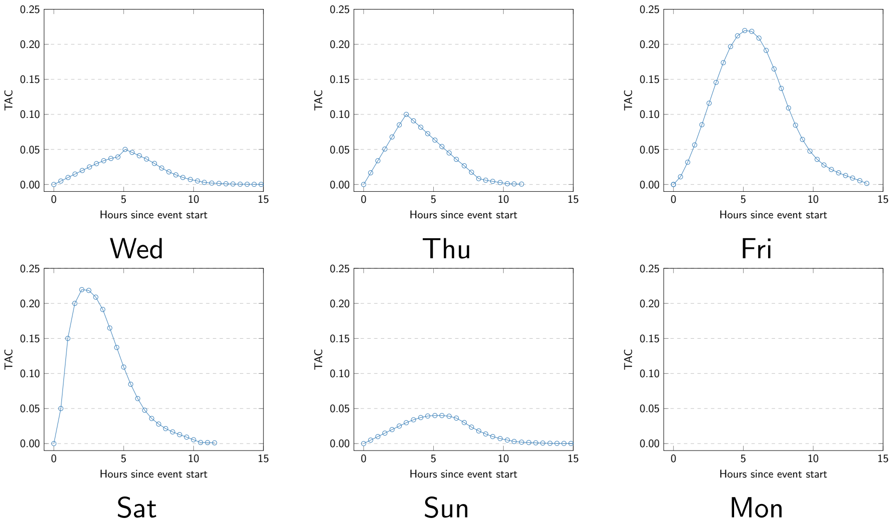
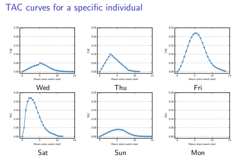
Study 1: Illustration Results



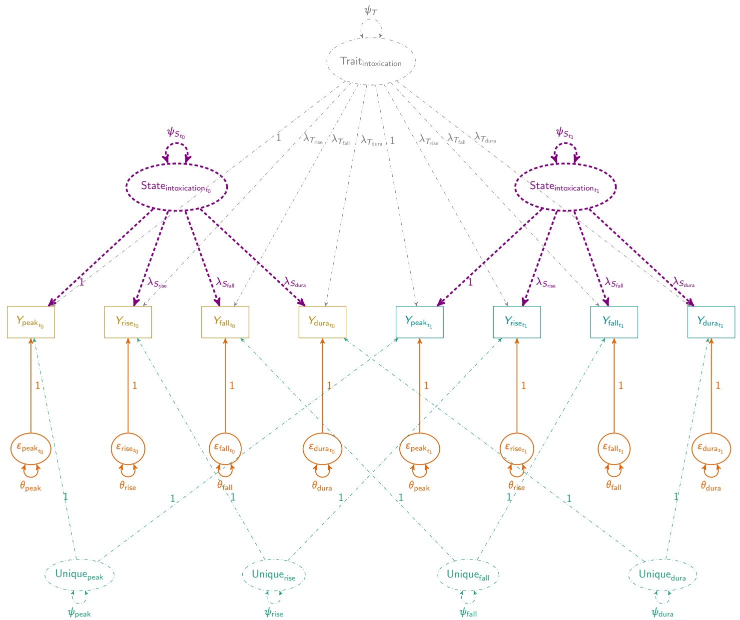


TAC curves for a specific individual





CUTS component of the CULTA model



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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

└─Extra slides

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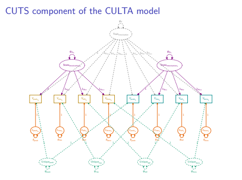
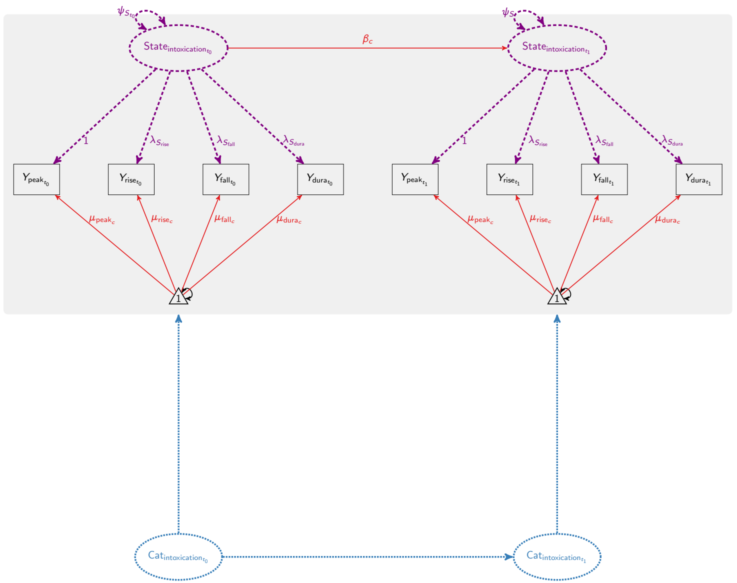
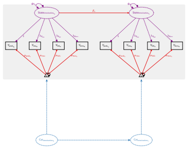
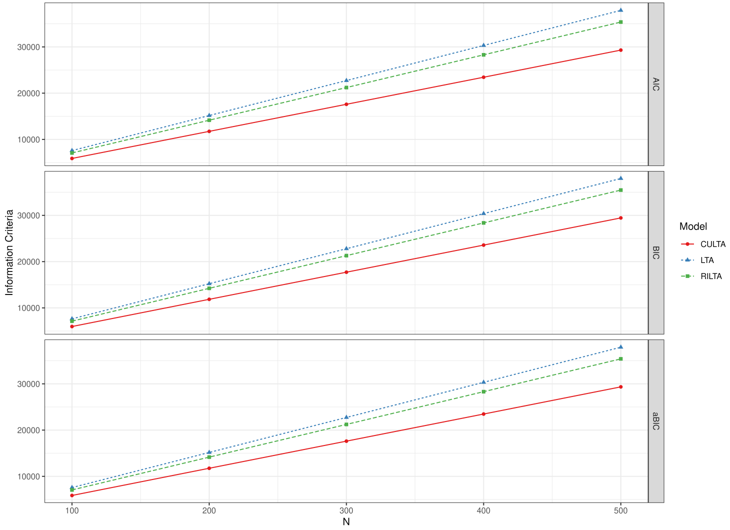


LTA component of the CULTA model





CULTA simulation: Information criteria



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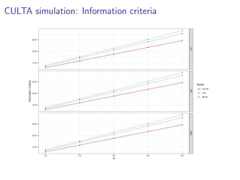
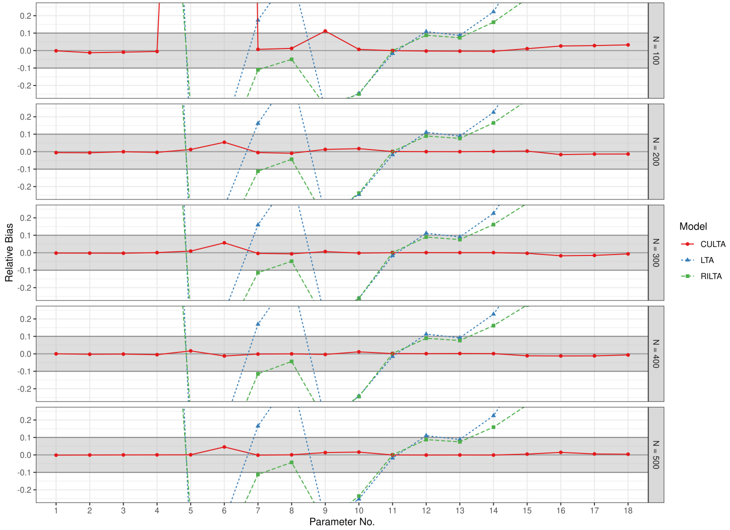


CULTA simulation: Relative bias



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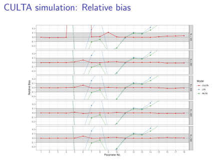
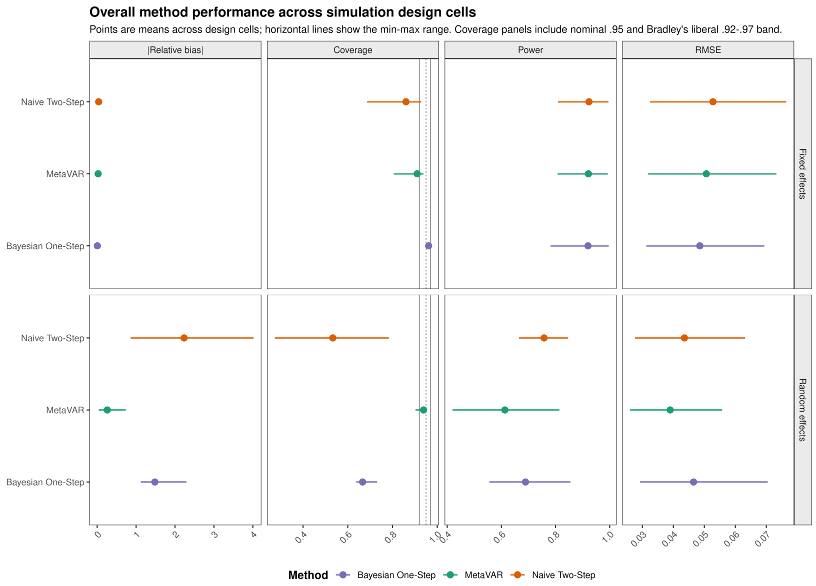
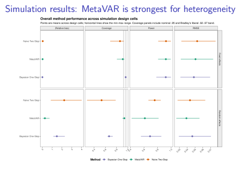
└Extra slides
└CULTA simulation: Relative bias



Simulation results: MetaVAR is strongest for heterogeneity





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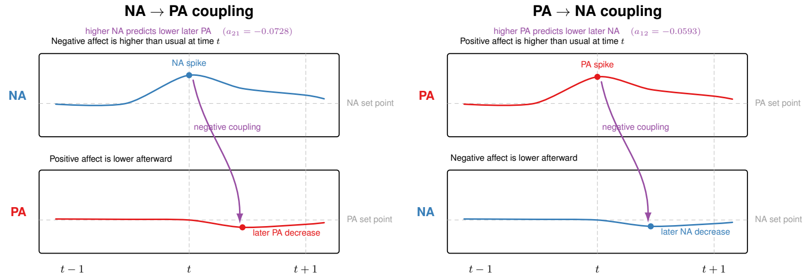
Empirical illustration: Coupling



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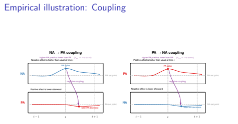
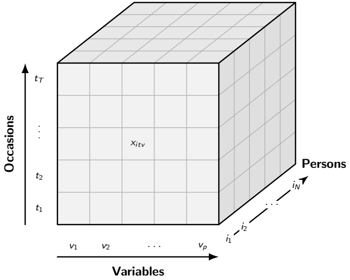
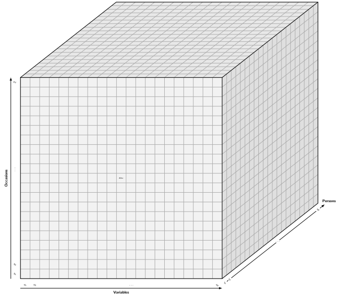
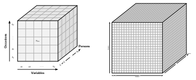


Cattell’s Box





Cattell’s Box



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